ASSIGNMENT REPORT-5

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Under the guidance of :

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For The Subject

Pattern Recognition and Machine Learning

(26AI5707)



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1. INTRODUCTION :

- The assignment focuses on estimating the parameters of Gaussian classes using maximum likelihood estimation (MLE).
- For Problem 1, three classes were generated using the Box–Muller polar method, following the Gaussian generation approach used in Assignment 4.
- Six covariance settings were tested: isotropic, diagonal and full covariance, with both identical and different covariance matrices across classes.
- Problem 2 applies the same MLE classification method to the provided two-dimensional real dataset.

2. EXPERIMENT STEPS :

A. GAUSSIAN DATA GENERATION :

- Three classes were generated with 10,000 samples per class, giving 30,000 samples in total.
- Standard Gaussian samples were generated with the Box–Muller polar method and transformed using the required means and covariance matrices.
- The class means used were $\mu_1 = [0, 0]$, $\mu_2 = [5, 5]$ and $\mu_3 = [2, -3]$.

B. COVARIANCE TYPES :

- Isotropic covariance uses the form $\sigma^2 I$, so both features have the same variance and zero covariance.
- Diagonal covariance allows different variances for the two features while keeping the off-diagonal values zero.
- Full covariance includes non-zero off-diagonal values, allowing the features to vary together.

C. TRAIN-TEST SPLIT :

- For Problem 1, 80% of the generated samples were used for training and 20% for testing.
- Training samples: 24,000; testing samples: 6,000.
- For Problem 2, the provided real dataset contained 300 samples, with 240 used for training and 60 for testing.

D. MAXIMUM LIKELIHOOD ESTIMATION :

- The mean vector and covariance matrix were estimated using only the training samples from each class.
- The covariance estimate used the maximum-likelihood denominator N rather than $N - 1$.
- Class prior probabilities were estimated from the proportion of training samples belonging to each class.

E. CLASSIFICATION AND EVALUATION :

- For each test point, the Gaussian discriminant score was calculated for every class, and the class with the highest score was selected.
- The decision-boundary plot shows the predicted class regions with test samples and estimated class means.
- The confusion matrix records correct and incorrect predictions for each class.
- ROC curves were plotted using a one-vs-rest approach, and the area under each curve (AUC) was reported.

3. PROBLEM 1 – MAXIMUM LIKELIHOOD ESTIMATION :

A. EXPERIMENT 1 – ISOTROPIC COVARIANCE – IDENTICAL :

- The covariance matrices used for Classes 1, 2 and 3 were: [[2, 0], [0, 2]]; [[2, 0], [0, 2]]; [[2, 0], [0, 2]].
- Total samples: 30,000; training samples: 24,000; testing samples: 6,000.

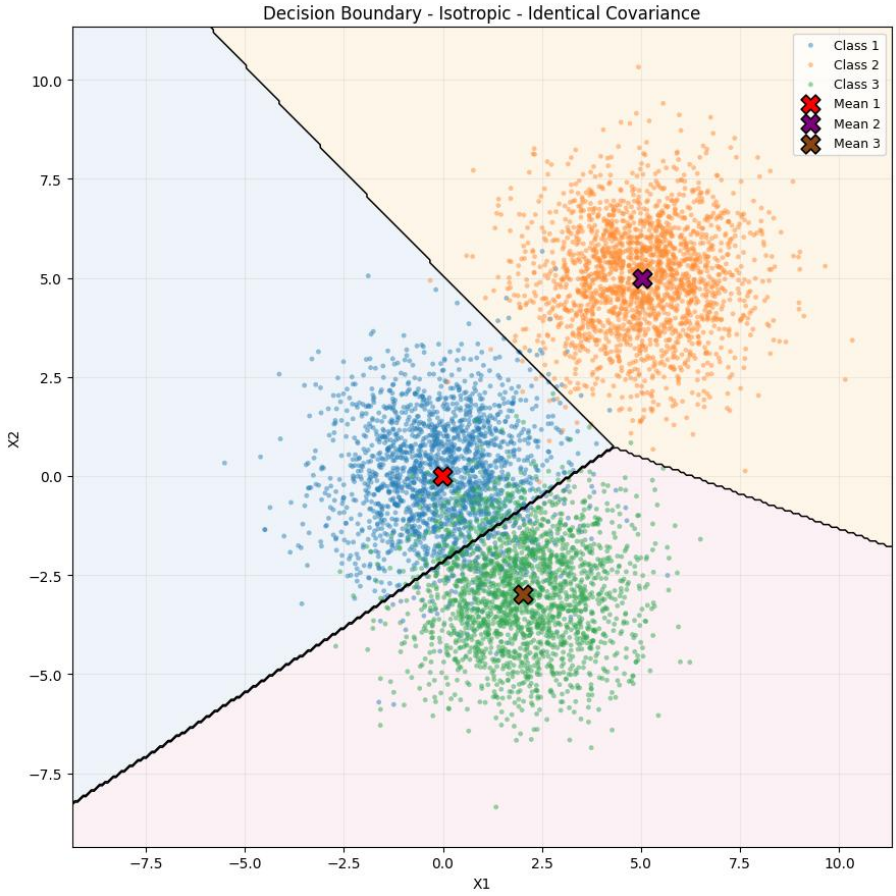


Figure 1 : Decision Boundary with Test Data – Experiment 1

- All classes use the same covariance matrix [[2, 0], [0, 2]].
- The estimated means are close to the means used to generate the data.
- The main classification errors occur between Class 1 and Class 3.
- Test accuracy = 92.43%.

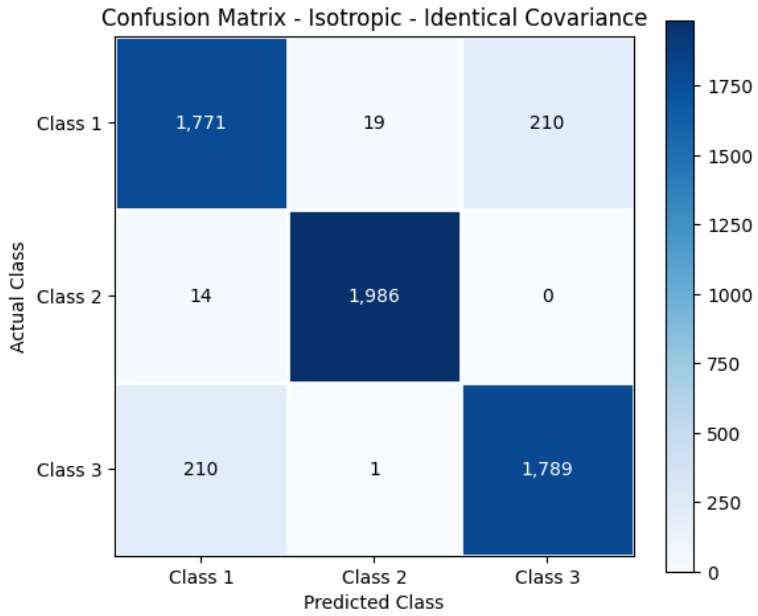


Figure 2 : Confusion Matrix – Experiment 1

Actual / Predicted	**Predicted C1**	**Predicted C2**	**Predicted C3**
Actual C1	1771	19	210
Actual C2	14	1986	0
Actual C3	210	1	1789

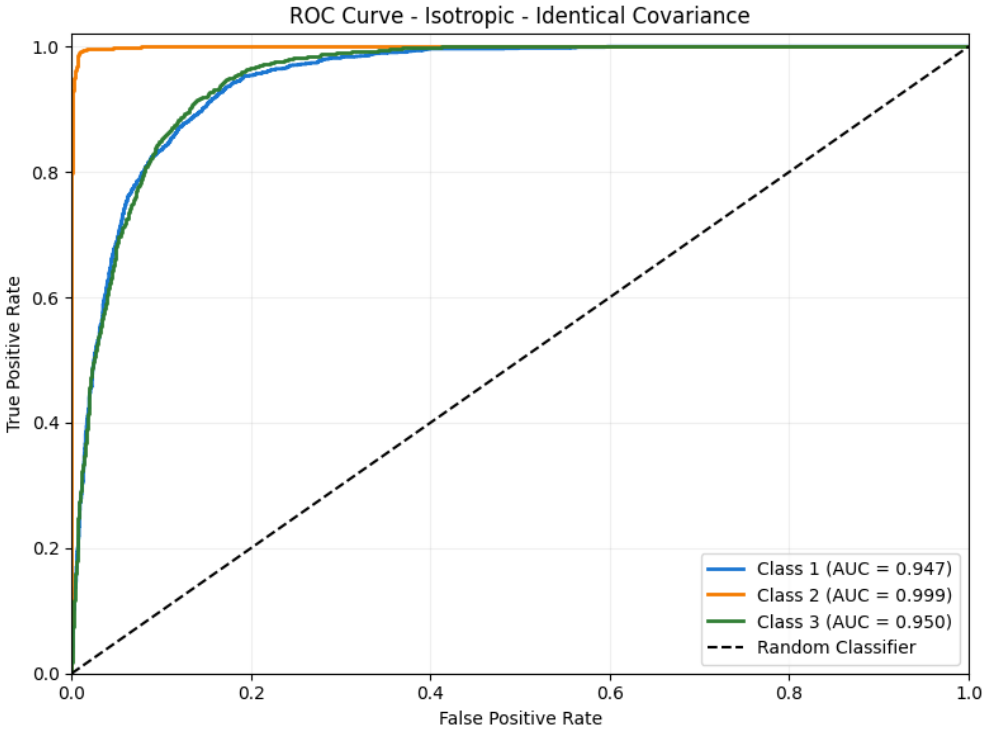


Figure 13 : ROC Curve – Experiment 1

- ROC-AUC values: Class 1 = 0.9473, Class 2 = 0.9993, Class 3 = 0.9503.

B. EXPERIMENT 2 – ISOTROPIC COVARIANCE – DIFFERENT :

- The covariance matrices used for Classes 1, 2 and 3 were: [[1, 0], [0, 1]]; [[2, 0], [0, 2]]; [[3, 0], [0, 3]].
- Total samples: 30,000; training samples: 24,000; testing samples: 6,000.

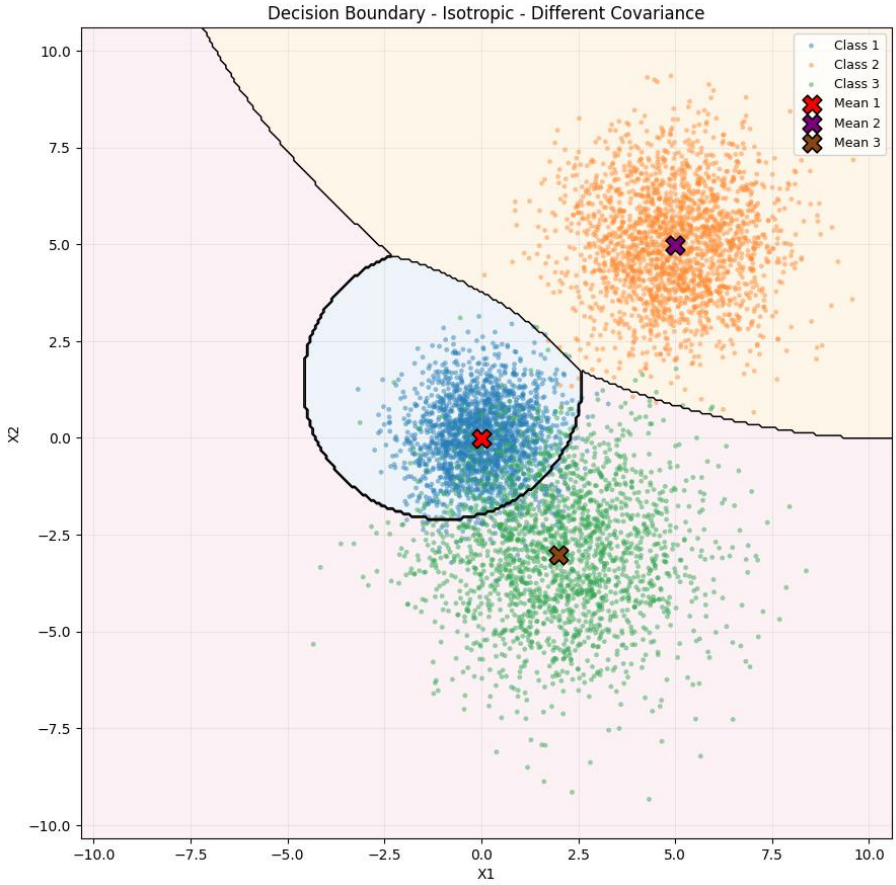


Figure 3 : Decision Boundary with Test Data – Experiment 2

- The three classes use different isotropic variances.
- Class 2 is classified with high accuracy; most errors involve Classes 1 and 3.
- The AUC for Class 2 is close to 1.0.
- Test accuracy = 94.27%.

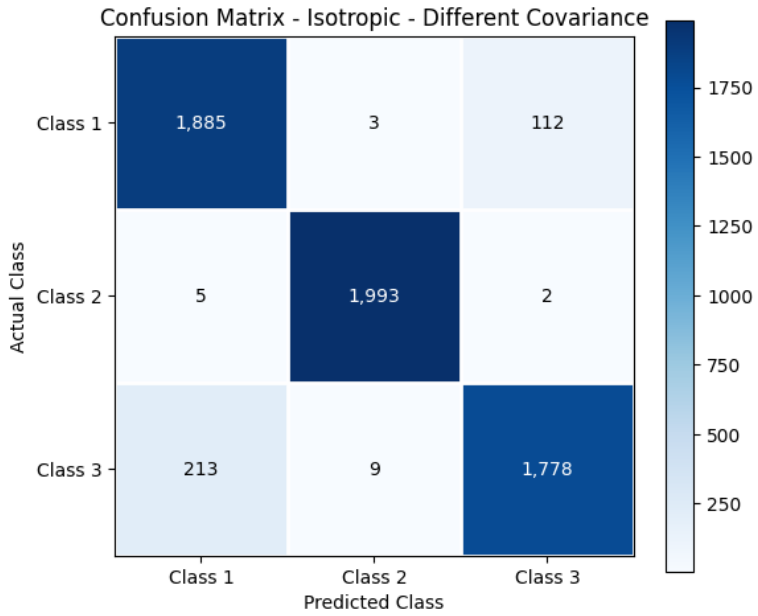


Figure 4 : Confusion Matrix – Experiment 2

Actual / Predicted	**Predicted C1**	**Predicted C2**	**Predicted C3**
Actual C1	1885	3	112
Actual C2	5	1993	2
Actual C3	213	9	1778

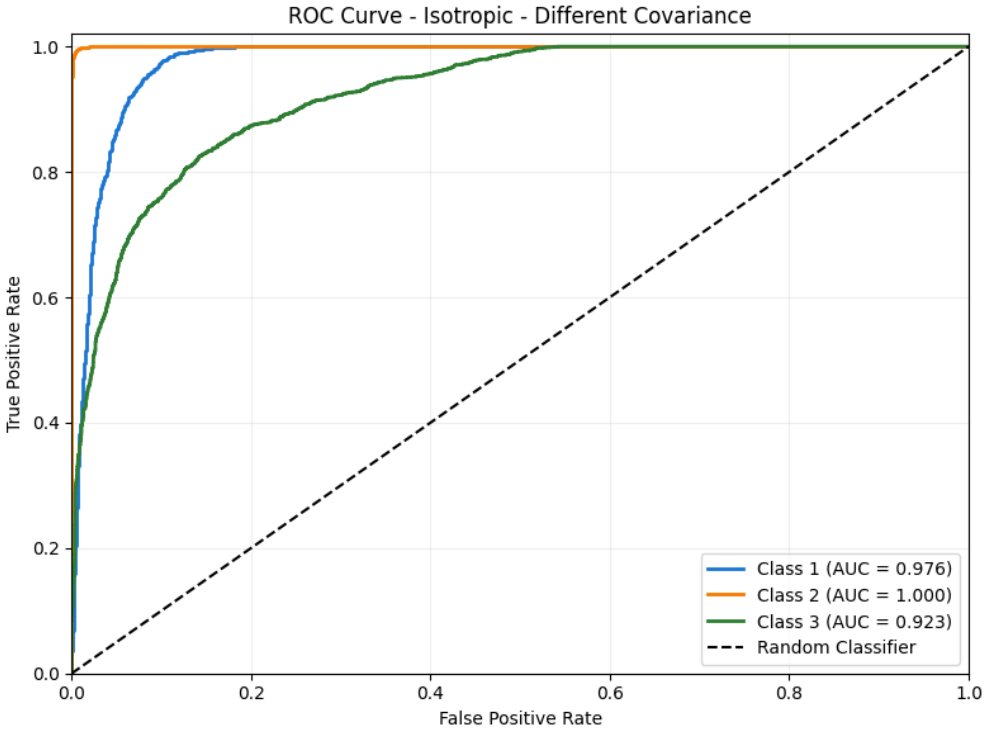


Figure 14 : ROC Curve – Experiment 2

- ROC-AUC values: Class 1 = 0.9764, Class 2 = 0.9999, Class 3 = 0.9234.

C. EXPERIMENT 3 – DIAGONAL COVARIANCE – IDENTICAL :

- The covariance matrices used for Classes 1, 2 and 3 were: [[2, 0], [0, 1]]; [[2, 0], [0, 1]]; [[2, 0], [0, 1]].
- Total samples: 30,000; training samples: 24,000; testing samples: 6,000.

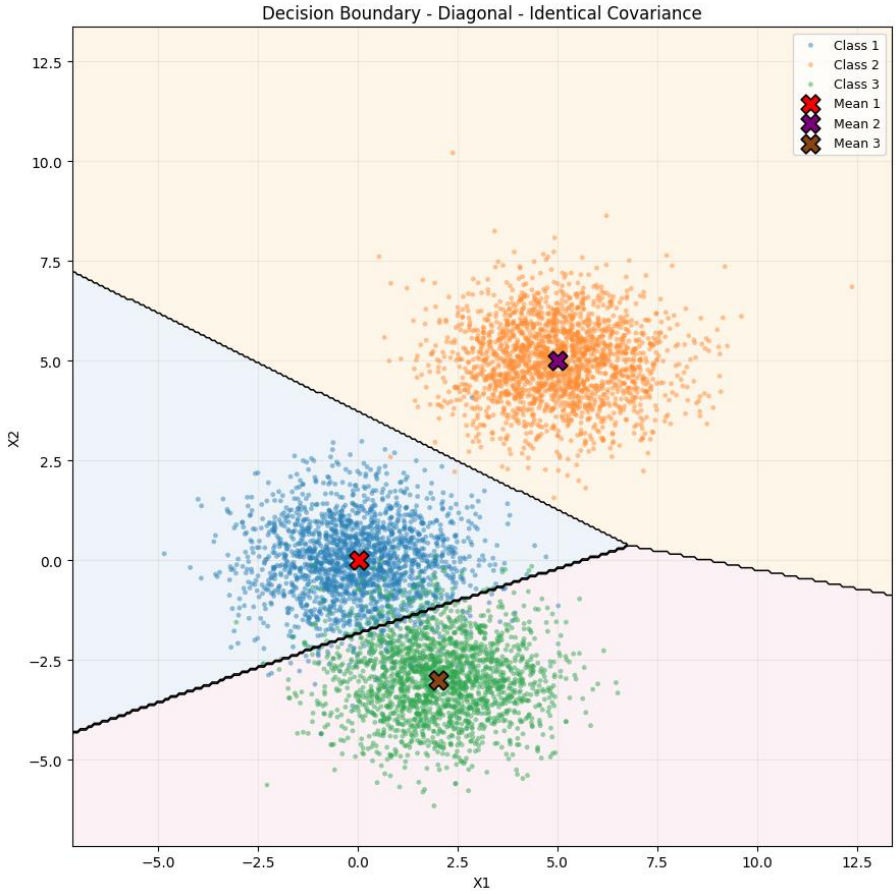


Figure 5 : Decision Boundary with Test Data – Experiment 3

- All classes use the same diagonal covariance matrix [[2, 0], [0, 1]].
- The two features have different variances, with zero off-diagonal covariance.
- This experiment gives the highest AUC values for Classes 1 and 3 among these six runs.
- Test accuracy = 96.20%.

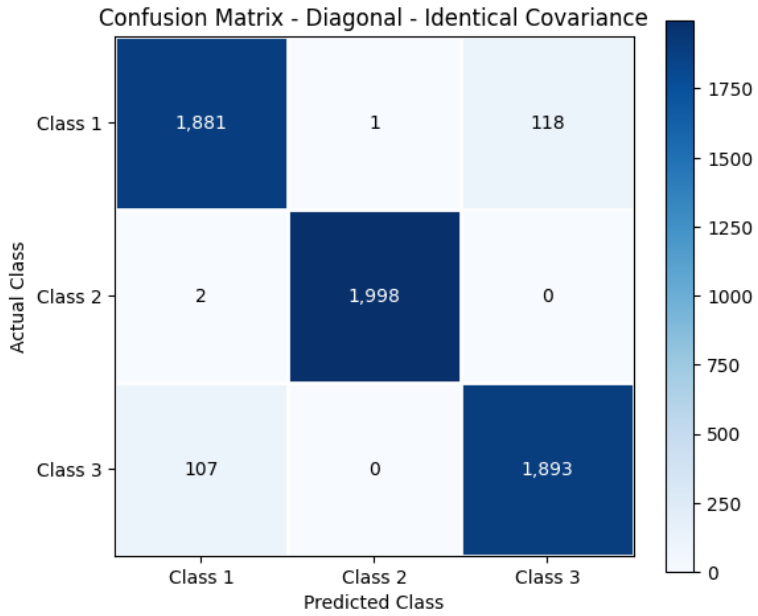


Figure 6 : Confusion Matrix – Experiment 3

Actual / Predicted	**Predicted C1**	**Predicted C2**	**Predicted C3**
Actual C1	1881	1	118
Actual C2	2	1998	0
Actual C3	107	0	1893

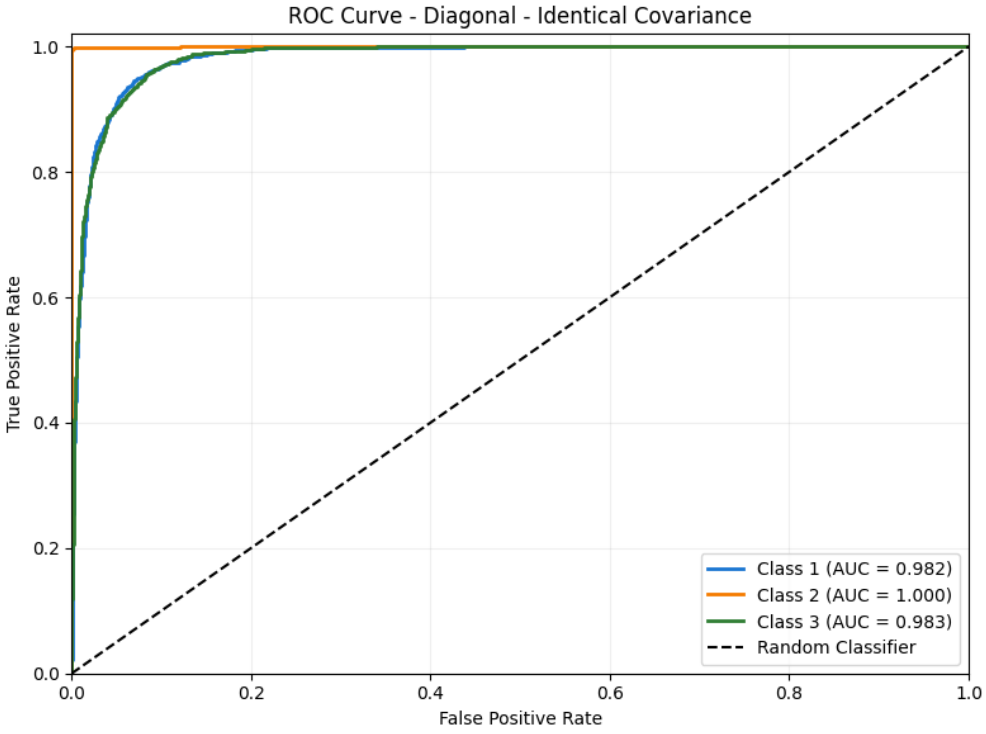


Figure 15 : ROC Curve – Experiment 3

- ROC-AUC values: Class 1 = 0.9825, Class 2 = 0.9998, Class 3 = 0.9827.

D. EXPERIMENT 4 – DIAGONAL COVARIANCE – DIFFERENT :

- The covariance matrices used for Classes 1, 2 and 3 were: [[1, 0], [0, 1]]; [[4, 0], [0, 1]]; [[2, 0], [0, 3]].
- Total samples: 30,000; training samples: 24,000; testing samples: 6,000.

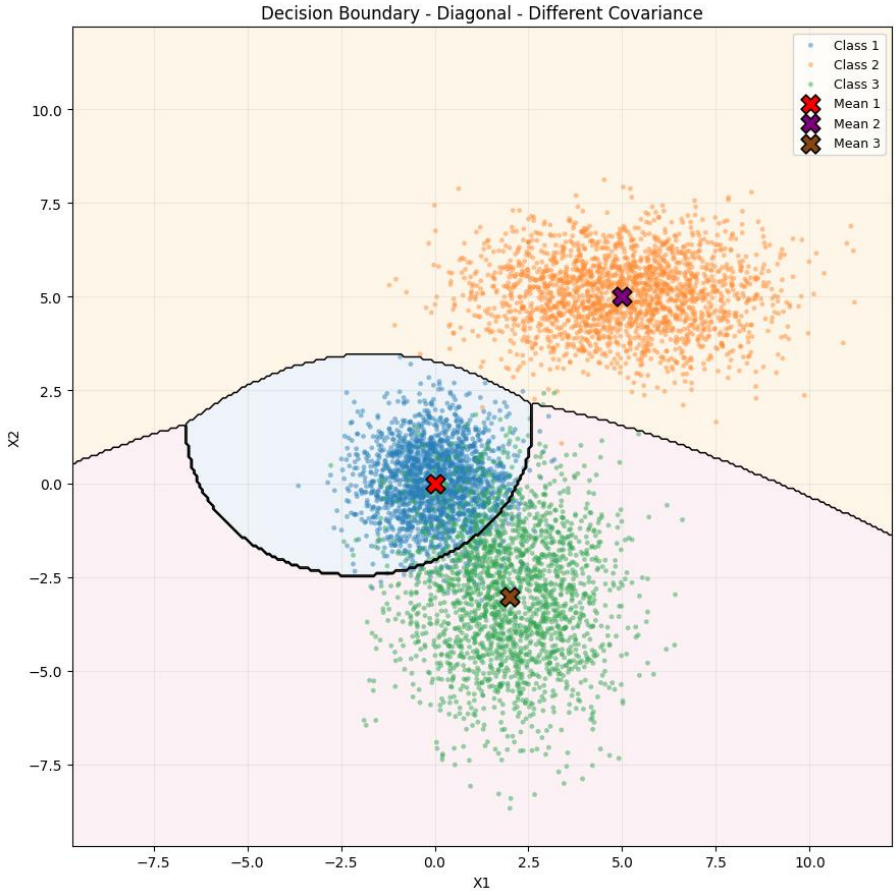


Figure 7 : Decision Boundary with Test Data – Experiment 4

- Each class has a different diagonal covariance matrix.
- Class 2 is separated well, while the main confusion occurs between Classes 1 and 3.
- The ROC-AUC for Class 2 is 1.0000 in this run.
- Test accuracy = 94.77%.

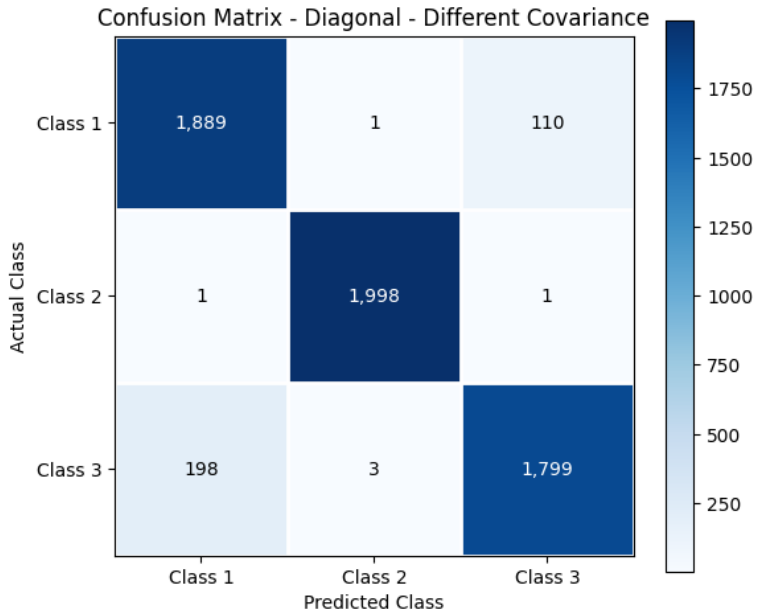


Figure 8 : Confusion Matrix – Experiment 4

Actual / Predicted	**Predicted C1**	**Predicted C2**	**Predicted C3**
Actual C1	1889	1	110
Actual C2	1	1998	1
Actual C3	198	3	1799

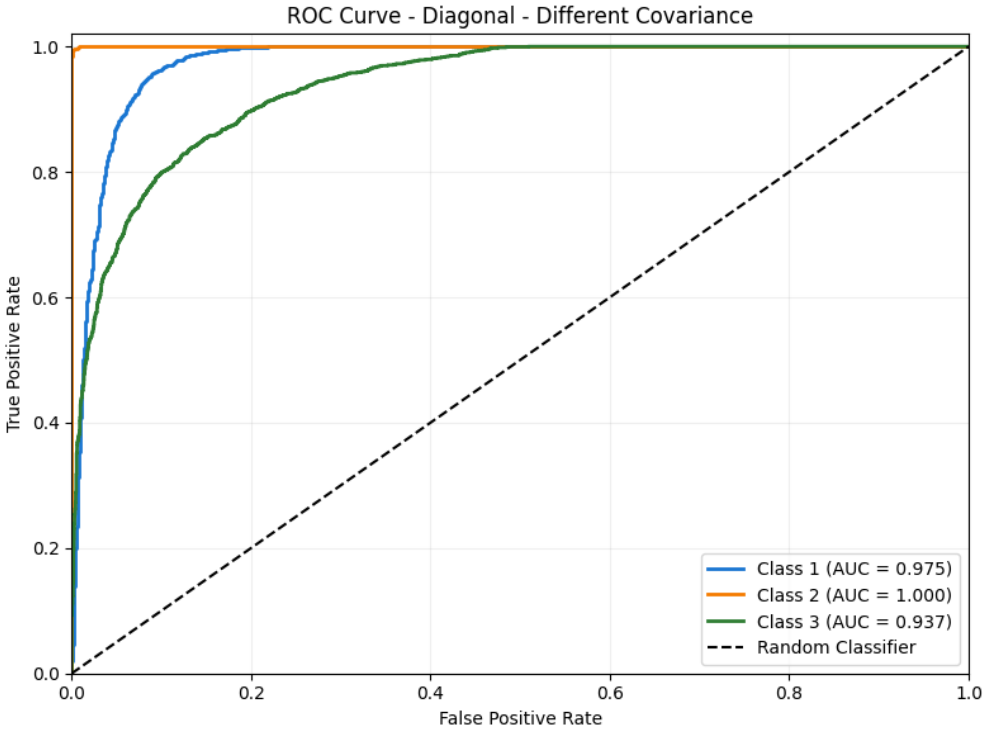


Figure 16 : ROC Curve – Experiment 4

- ROC-AUC values: Class 1 = 0.9754, Class 2 = 1.0000, Class 3 = 0.9372.

E. EXPERIMENT 5 – FULL COVARIANCE – IDENTICAL :

- The covariance matrices used for Classes 1, 2 and 3 were: [[2, 1], [1, 2]]; [[2, 1], [1, 2]]; [[2, 1], [1, 2]].
- Total samples: 30,000; training samples: 24,000; testing samples: 6,000.

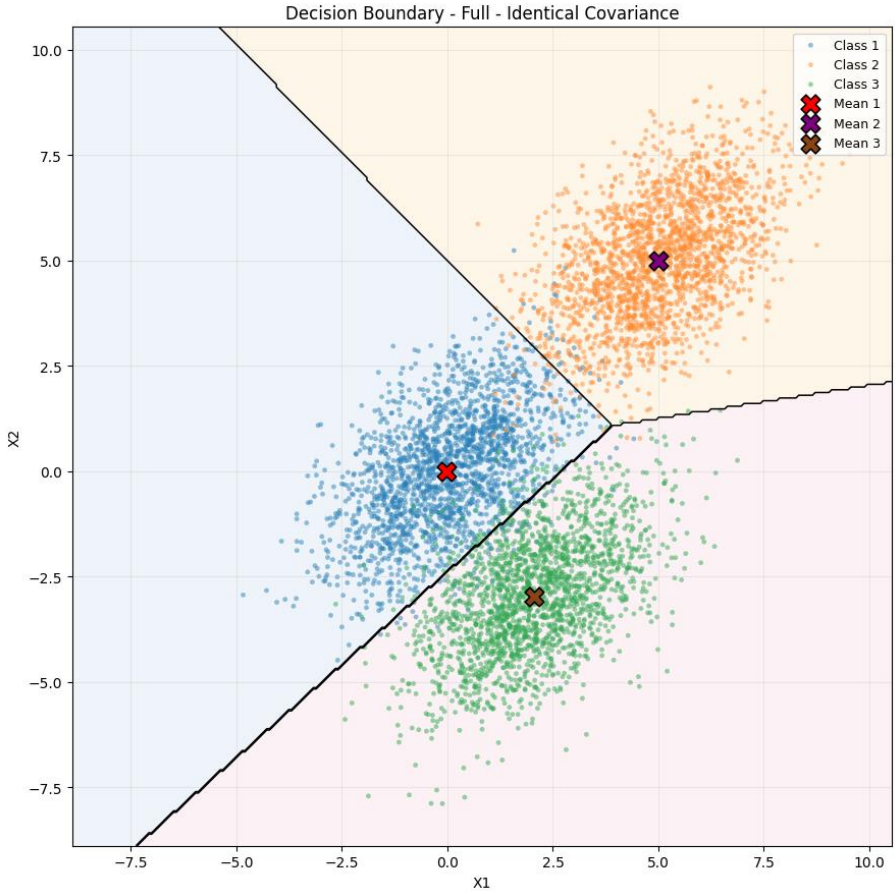


Figure 9 : Decision Boundary with Test Data – Experiment 5

- The same full covariance matrix [[2, 1], [1, 2]] is used for all classes.
- The non-zero off-diagonal values represent covariance between the two features.
- This run has the highest test accuracy among the six experiments.
- Test accuracy = 96.40%.

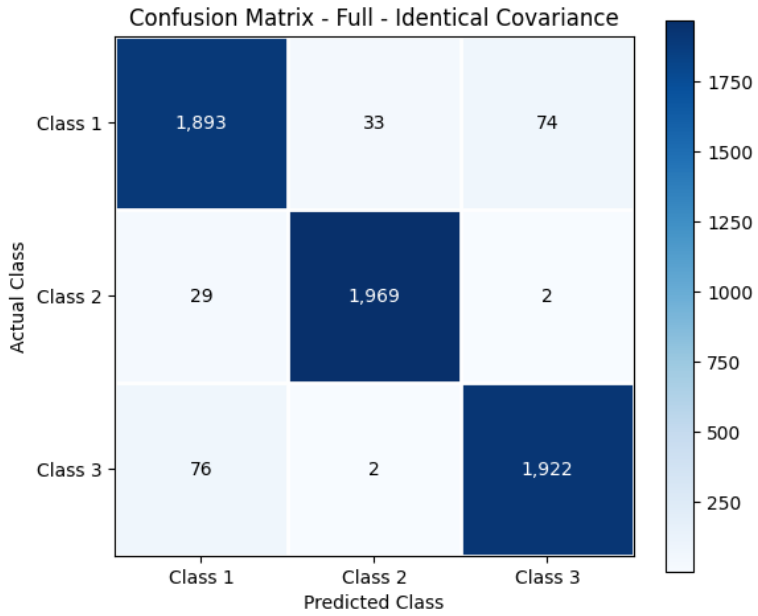


Figure 10 : Confusion Matrix – Experiment 5

Actual / Predicted	**Predicted C1**	**Predicted C2**	**Predicted C3**
Actual C1	1893	33	74
Actual C2	29	1969	2
Actual C3	76	2	1922

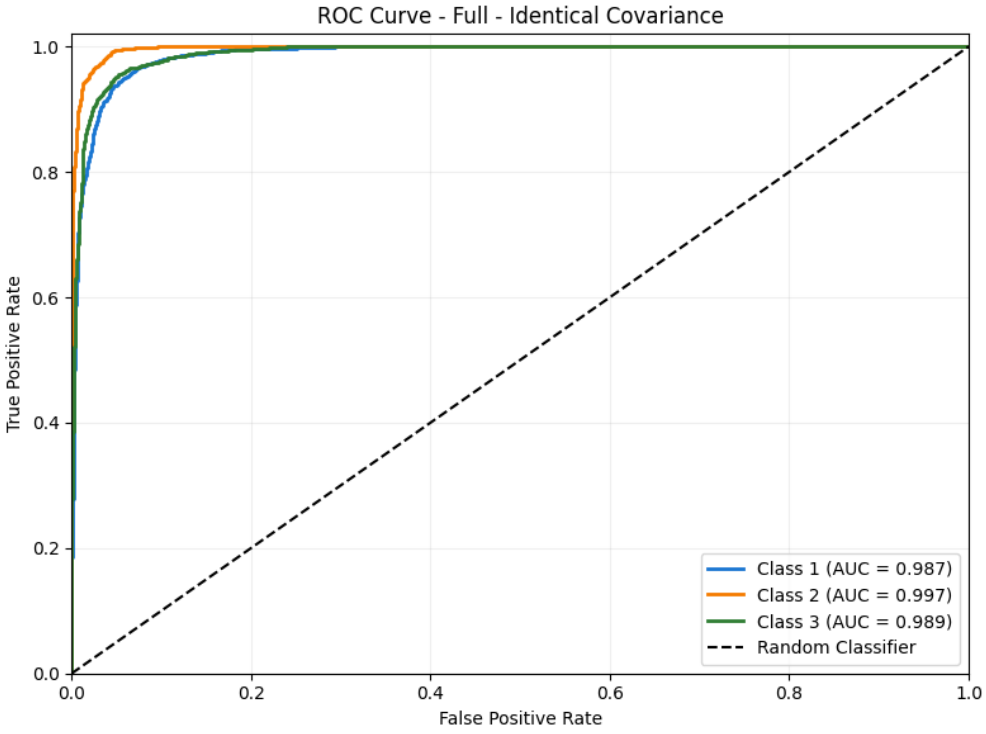


Figure 17 : ROC Curve – Experiment 5

- ROC-AUC values: Class 1 = 0.9870, Class 2 = 0.9968, Class 3 = 0.9890.

F. EXPERIMENT 6 – FULL COVARIANCE – DIFFERENT :

- The covariance matrices used for Classes 1, 2 and 3 were: [[2, 1], [1, 2]]; [[2, -1], [-1, 2]]; [[3, 0.7], [0.7, 1.5]].
- Total samples: 30,000; training samples: 24,000; testing samples: 6,000.

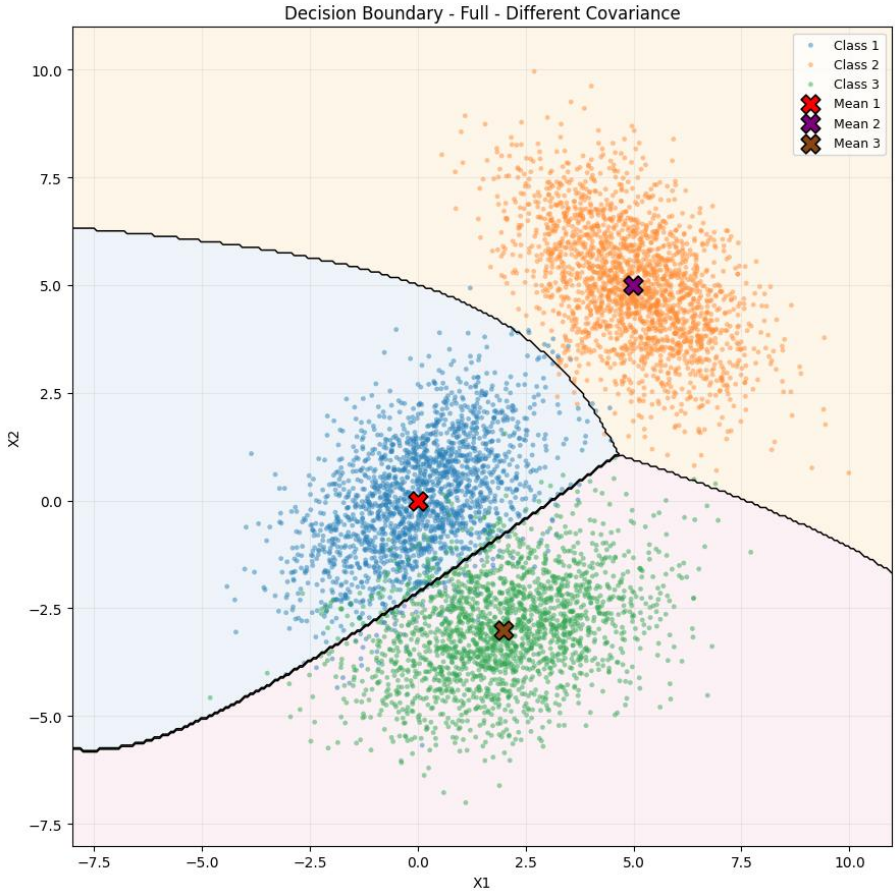


Figure 11 : Decision Boundary with Test Data – Experiment 6

- Different full covariance matrices are used for the three classes.
- The covariance matrices have different spreads and covariance directions.
- The test accuracy is close to the highest result among the six experiments.
- Test accuracy = 96.38%.

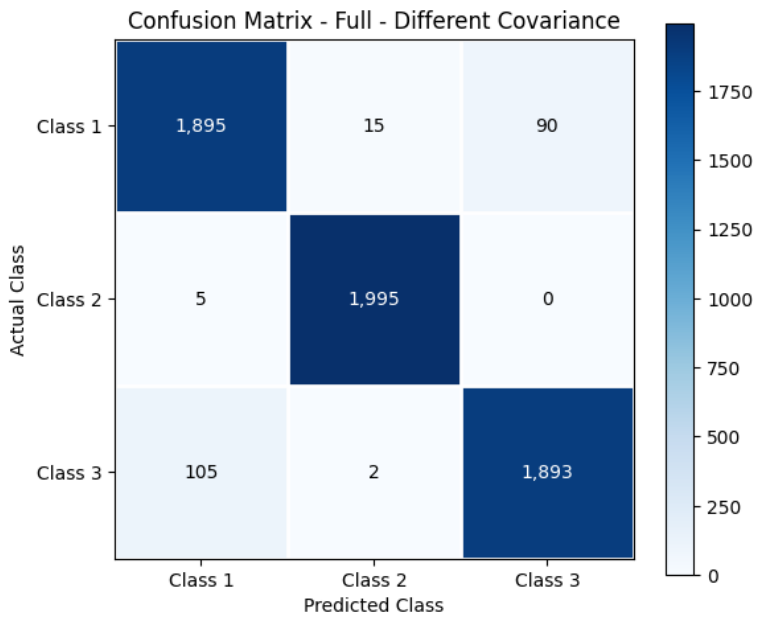


Figure 12 : Confusion Matrix – Experiment 6

Actual / Predicted	**Predicted C1**	**Predicted C2**	**Predicted C3**
Actual C1	1895	15	90
Actual C2	5	1995	0
Actual C3	105	2	1893

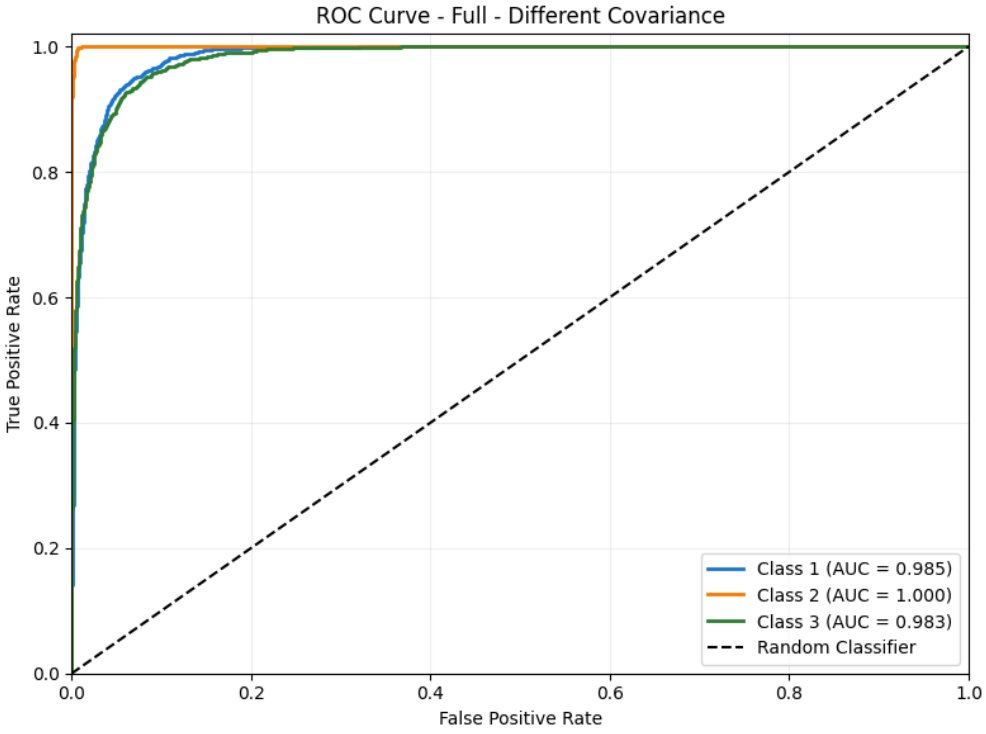


Figure 18 : ROC Curve – Experiment 6

- ROC-AUC values: Class 1 = 0.9849, Class 2 = 0.9998, Class 3 = 0.9835.

4. RESULT COMPARISON :

Covariance Type	Condition	Test Accuracy	Test Samples	Mean AUC
Isotropic	Identical	92.43%	6,000	0.9656
Isotropic	Different	94.27%	6,000	0.9666
Diagonal	Identical	96.20%	6,000	0.9883
Diagonal	Different	94.77%	6,000	0.9709
Full	Identical	96.40%	6,000	0.9909
Full	Different	96.38%	6,000	0.9894

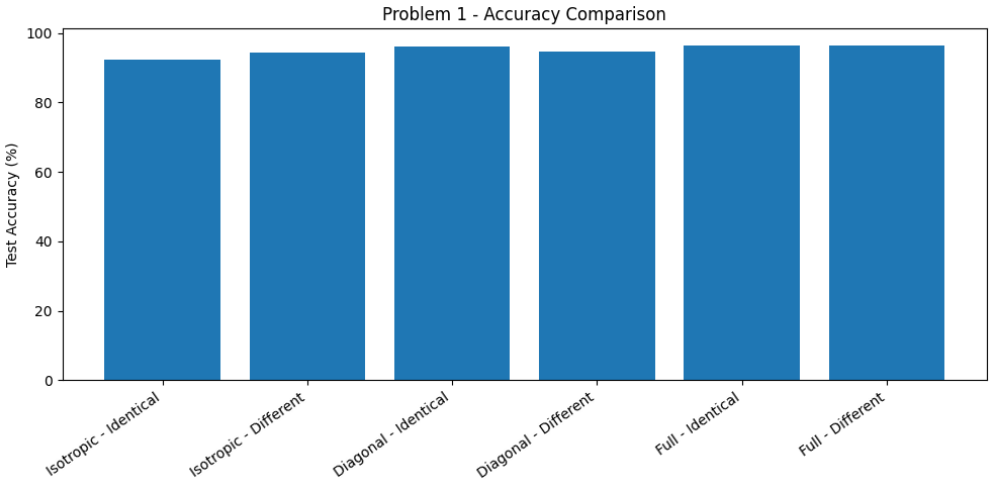


Figure 19 : Comparison of MLE Classification Accuracy

- The full covariance with identical matrices gives the highest test accuracy in this run, at 96.40%.
- The full covariance with different matrices gives 96.38%, which is very close to the highest result.
- The isotropic identical case has the lowest accuracy in this run, at 92.43%.
- Class 2 has consistently high AUC values across the six experiments.

5. PROBLEM 2 – REAL DATASET CLASSIFICATION :

A. DATASET AND TRAIN-TEST SPLIT :

- The supplied two-dimensional dataset contains 300 samples and three class labels.
- The dataset was split into 240 training samples and 60 test samples using an 80/20 split.
- The same Gaussian MLE classifier was applied to the real data.

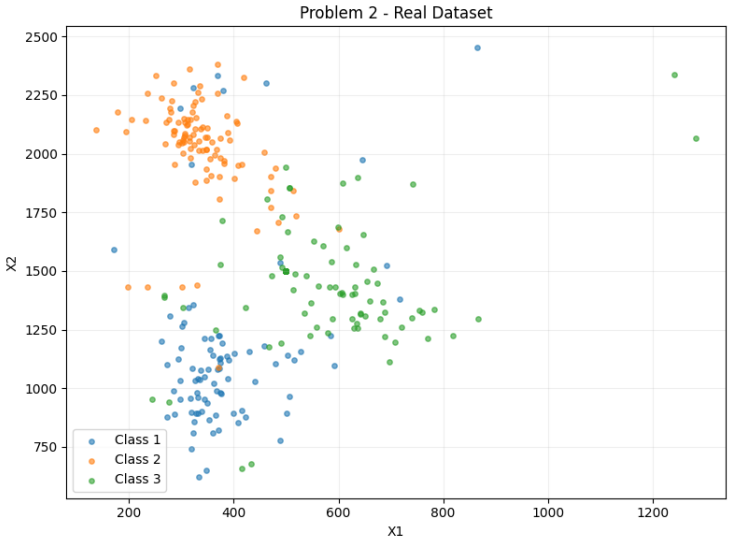


Figure 20 : Scatter Plot of the Real Dataset

B. MAXIMUM LIKELIHOOD ESTIMATION :

- Class 1: estimated mean = [395.3066, 1211.0428].
- Class 1: estimated covariance = [[11666.2500, 16763.0831], [16763.0831, 137120.8953]].
- Class 2: estimated mean = [341.7579, 2026.1512].
- Class 2: estimated covariance = [[6116.3049, -4458.4330], [-4458.4330, 40737.9815]].
- Class 3: estimated mean = [564.3743, 1442.1571].
- Class 3: estimated covariance = [[19802.5270, 3867.6668], [3867.6668, 45076.3070]].

C. DECISION BOUNDARY :

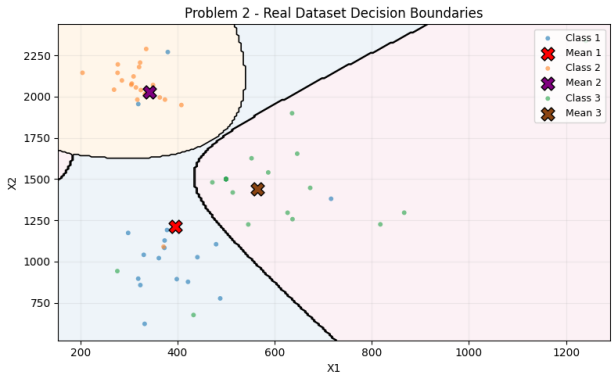


Figure 21 : Decision Boundaries for the Real Dataset

- The decision regions are based on the estimated Gaussian parameters. The axes use separate feature ranges because X1 and X2 have substantially different numerical ranges.

D. CONFUSION MATRIX AND TEST ACCURACY :

- Test accuracy = 85.00% (51 correctly classified samples out of 60).

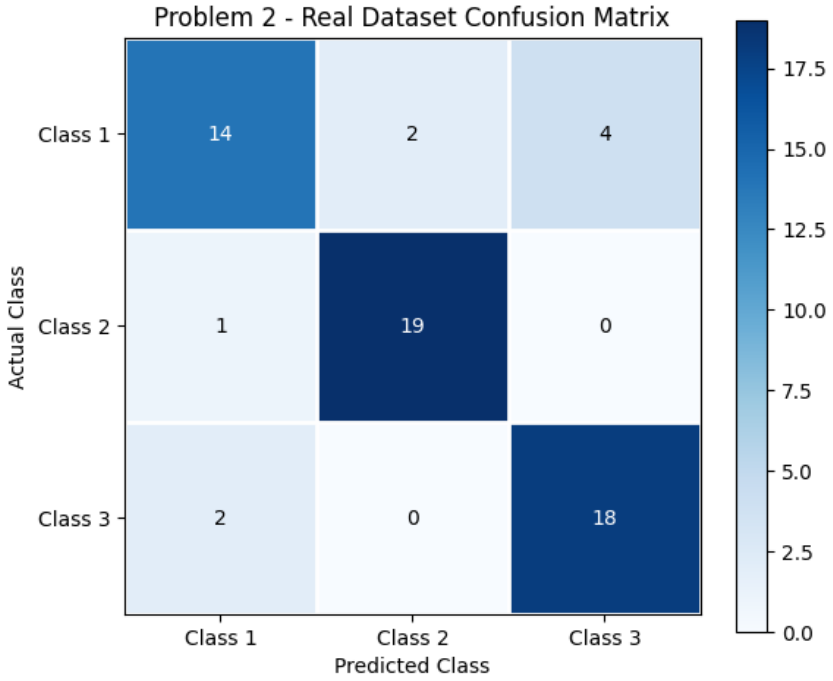


Figure 22 : Confusion Matrix for the Real Dataset

Actual / Predicted	Predicted C1	Predicted C2	Predicted C3
Actual C1	14	2	4
Actual C2	1	19	0
Actual C3	2	0	18

- Class 1 has six incorrect predictions: two were assigned to Class 2 and four to Class 3.
- Class 2 has 19 correct predictions, with one sample classified as Class 1.
- Class 3 has 18 correct predictions and two samples classified as Class 1.

E. ROC CURVE :

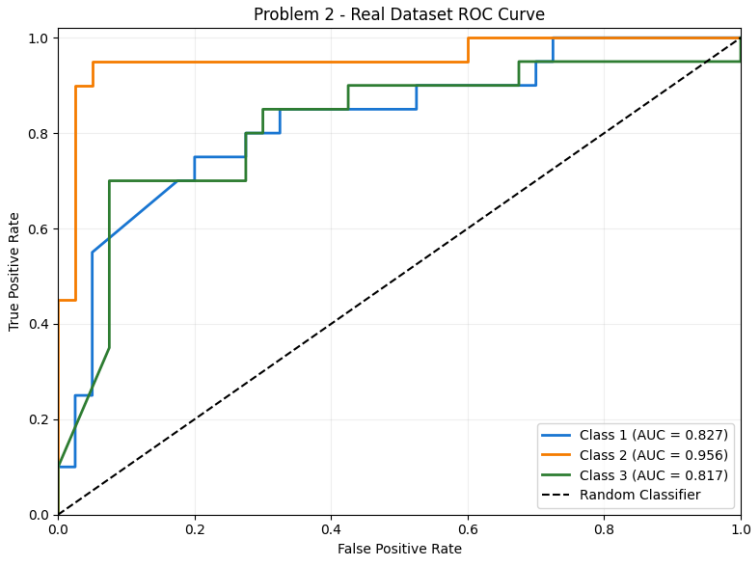


Figure 23 : ROC Curve for the Real Dataset

- ROC-AUC values: Class 1 = 0.8269, Class 2 = 0.9562 and Class 3 = 0.8169.
- Class 2 has the highest AUC, indicating the strongest one-vs-rest separation in this test set.

6. INFERENCE :

- The generated-data experiments show that the covariance structure affects the shape of the decision regions and the final classification accuracy.
- For the current run, the full covariance with identical matrices achieved the highest Problem 1 test accuracy of 96.40%, closely followed by the full covariance with different matrices at 96.38%.
- The confusion matrices show that most generated test samples were classified correctly, although confusion between Classes 1 and 3 remains in several settings.
- The ROC curves provide a class-wise view of performance that is not captured by accuracy alone.
- For Problem 2, the real dataset achieved 85.00% test accuracy. Class 2 had the highest AUC (0.9562), while Classes 1 and 3 had AUC values of 0.8269 and 0.8169.
- Overall, MLE provided a straightforward way to estimate Gaussian class parameters and classify both generated and real two-dimensional data.